

Table 7.4 Monthly Volume Forecast Regression Results

Large Cap Stocks (SP500)						Small Cap Stocks (SP500)					
Ten Years: 2002–2011						Ten Years: 2002–2011					
	const	$\Delta V(-1)$	$\Delta V(-12)$	$\Delta\sigma$	ΔS_{px}		const	$\Delta V(-1)$	$\Delta V(-12)$	$\Delta\sigma$	ΔS_{px}
beta	0.0056	-0.2990	0.1354	0.5461	-0.1877	beta	0.0029	-0.2212	0.2385	0.3569	0.3437
se	0.0064	0.0435	0.0461	0.0328	0.1466	se	0.0077	0.0732	0.0784	0.0475	0.1717
t-stat	0.8806	-6.8785	2.9352	16.6699	-1.2798	t-stat	0.3773	-3.0216	3.0423	7.5084	2.0016
R ²	0.79					R ²	0.39				
Recent Five Years: 2007–2011						Recent Five Years: 2007–2011					
	const	$\Delta V(-1)$	$\Delta V(-12)$	$\Delta\sigma$	ΔS_{px}		const	$\Delta V(-1)$	$\Delta V(-12)$	$\Delta\sigma$	ΔS_{px}
beta	-0.001	-0.313	0.088	0.529	-0.432	beta	-0.001	-0.225	0.143	0.362	0.140
se	0.009	0.060	0.061	0.042	0.196	se	0.012	0.100	0.109	0.058	0.222
t-stat	-0.072	-5.245	1.461	12.601	-2.201	t-stat	-0.103	-2.254	1.310	6.259	0.629
R ²	0.83					R ²	0.49				

We estimated our regression coefficients for large and small cap stocks using a ten year horizon. We further defined three periods to evaluate the stability of these relationships. These periods are:

Ten Years: 1/2002–12/2011

Five Years: 1/2007–12/2011

Five Years: 1/2002–12/2006

Regression results:

The result of our regression study is shown in [Table 7.4](#). In total, there are six scenarios that were analyzed—three for LC and three for SC. The results show the estimated betas, corresponding standard errors and t-stat, and the R^2 statistic. Overall our regression model had a very strong fit. The model did explain a larger percentage of the variation for large cap stocks than for small cap stocks (as is expected due to the trading stability of the smaller companies). The R^2 using 10 years of data was $R^2 = 0.70$ for large cap stocks and $R^2 = 0.39$ for small cap stocks. Overall, our formulation is a very statistically sound model.